

PyGPT

A Python code-completion assistant, trained from scratch

Every stage built and measured by hand: data → tokenizer → model → training → evaluation → deployment

Rehan · Machine Learning · Bennett University · October 2026

The problem and the end goal

- Task: given the start of a Python function, predict what comes next — next-token prediction is code completion
- Goal: a working assistant powered by a model we trained ourselves — not an API call, not a fine-tune
- “Working” means three things:
 - a live demo that completes code in seconds
 - measured results against baselines (perplexity, syntactic validity, functional tests)
 - a public repo whose structure matches the design in this deck
- Constraints: 3 days, free compute only (Kaggle 2× T4, 30 GPU-h/week), solo

Existing systems — and the gap

Criterion 1

System	Year	Params	Open?	Relevance
GPT-2 (Radford et al.)	2019	124M–1.5B	Yes	Architecture we reuse; English-only → our baseline
Codex / Copilot (Chen et al.)	2021	12B	No	Defined the task and pass@k evaluation
CodeParrot (HF)	2021	110M / 1.5B	Yes	Closest prior work: same data family, 50–100× our compute
SantaCoder / StarCoder (BigCode)	2022–23	1.1B / 15.5B	Yes	License-filtered data pipeline we follow in spirit
Code Llama (Meta)	2023	7B–70B	Yes	Open state of the art
PyGPT (ours)	2026	110.5M	Yes	Complete, reproducible, measured pipeline at student scale

Gap: everything above is closed-source or needs 100–10,000× a student's compute. None of it teaches how a code LM is built. PyGPT's contribution is honest numbers on what ~7 GPU-hours buys — with every stage visible.

Feasibility: do the math before burning GPU hours

Criterion 1

- Scaling-law estimate: $\text{FLOPs} \approx 6 \cdot N \cdot D = 6 \times 1.1\text{e}8 \times 8\text{e}8 \approx 5.3\text{e}17 \rightarrow \sim 6 \text{ h on } 2 \times \text{T4 at } 20\text{--}30\% \text{ utilization}$
- Measured on the day: 34,000 tok/s (fp16, DDP) $\rightarrow$ 799.5M tokens in 6 h 45 min — one full epoch
- Chinchilla-optimal for 110M params $\approx$ 2.2B tokens $\rightarrow$ we are $\sim 3\times$ under-trained by design (limitations slide)
- The free-tier landscape is a moving target — verify, don't assume:
 - AWS: GPU quota denied twice (new account) $\rightarrow$ dead end
 - Hugging Face: free Gradio Spaces discontinued mid-project (402) $\rightarrow$ laptop demo instead
 - Kaggle: 30 GPU-h/week free — 16.5 h spent incl. a 2nd epoch and a DSA curriculum phase; 13.5 h reserve never needed
- Session-death insurance: checkpoint every 250 steps + tested kill-and-resume before the long run

Objectives — all measurable, all measured

Criterion 2

#	Objective	Target	Achieved
O1	GPT-2-class model from scratch in PyTorch	init loss $\approx \ln(32768)=10.4$; causal-mask test	10.42 ✓ · mask test ✓ · 110.5M params ✓
O2	Clean Python pretraining corpus	$\geq 800\text{M}$ train tokens + held-out val	800M + 5M (codeparrot-clean) ✓
O3	Useful loss on free GPUs	val loss < 1.8 in ≤ 10 GPU-h	1.639 after 2 epochs ✓ · shipped DSA-specialized model: 1.82 (slide 9)
O4	Quantitative eval vs baselines	ppl, AST-validity, pass@k	AST-valid 57% vs GPT-2 7% ✓ · DSA pass@10 0.58 vs GPT-2 0.00 ✓
O5	Deployed completion app	30-token prompt answered < 15 s	2.4 s for 96 tokens on laptop CPU ✓
O6	Design $\leftrightarrow$ implementation sync	map table + public repo	slide 11 + GitHub + hf.co/rehannn11223/pygpt ✓
O7	(stretch) VS Code extension	60-s video	future work

Methodology: the pipeline

Criterion 2 — every box is a file in the repo

```
HF Hub: codeparrot-clean (streamed, never fully downloaded)
      | filters: size, line length, %alpha, auto-generated
      ▼
data/prepare_data.py — BPE tokenize (32k code vocab) — <|endoftext|> joins files
      ▼
train.bin / val.bin (uint16 mmap — vocab 32,768 fits exactly)
      ▼
train.py  AdamW · warmup+cosine · fp16+GradScaler · grad-accum · DDP on 2× T4
      |      checkpoints every 250 steps · resume · CSV log
      ▼
ckpt_best.pt → eval.py  val ppl · AST-validity · pass@k · baselines
      ▼
export.py  fp16 weights-only (212 MB) → app/app.py (Gradio, runs on a laptop CPU)
```

Architecture: GPT-2 Small, written from scratch

Criteria 3+4 — file and class names on every box

```
tokens → wte (32768×768) + wpe (512×768)      model.py: GPT
      ▼
12 × Block:  x = x + CausalSelfAttention(LN(x))  model.py: Block, CausalSelfAttention
             x = x + MLP(LN(x))                 model.py: MLP    (768 → 3072 → 768, GELU)
      ▼
LayerNorm → lm_head (tied to wte)              weight tying: -25M params
```

- Parameters: embeddings 25.6M + 12 blocks × 7.1M ≈ 110.5M (“124M-class”: GPT-2's 50k English vocab vs our 32k code vocab)
- Why each choice: pre-LN → stable at 12 layers · SDPA kernel → memory-efficient attention on T4 · ctx 512 → 1.4× faster, functions fit
- Generation: temperature / top-k / nucleus sampling + per-layer KV cache (each step feeds 1 token, not the whole sequence)

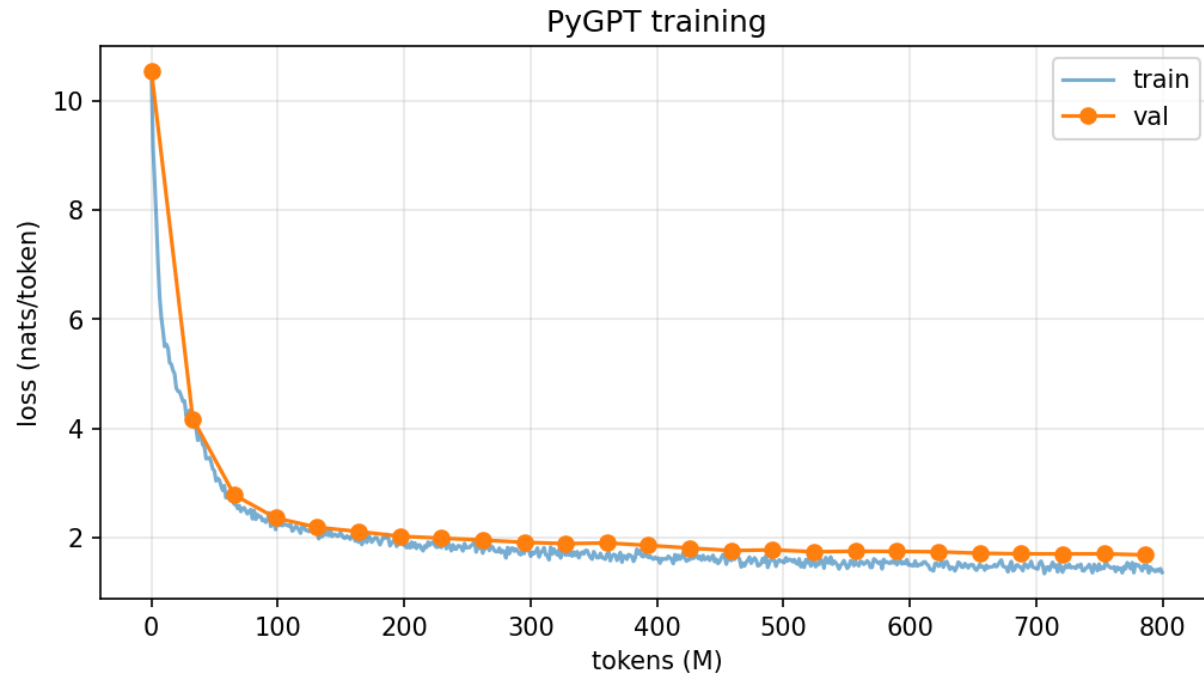
Data, tokenizer, training recipe — each row has a why

Criterion 3

Choice	Value	Why
Data	codeparrot-clean, 800M tokens streamed	pre-deduplicated Python; streaming avoids a 50 GB download
Tokenizer	codeparrot BPE, vocab 32,768	code-trained: indentation + identifiers are single tokens (~35% fewer than GPT-2's)
Optimizer	AdamW $\beta=(0.9, 0.95)$, wd 0.1 on matrices only	decoupled decay regularizes without fighting Adam
LR schedule	warmup 200 $\rightarrow$ 6e-4 $\rightarrow$ cosine $\rightarrow$ 6e-5	warmup avoids early divergence; low final LR sharpens the minimum
Precision	fp16 autocast + GradScaler	2–3 $\times$ throughput on T4 tensor cores; scaler prevents underflow
Batching	16 $\times$ 512 $\times$ 2 GPUs $\times$ grad-accum 8 = 131k tok/step	large effective batch within 16 GB
Multi-GPU	DistributedDataParallel (torchrun)	near-linear scaling on the 2 free T4s
Safety	grad clip 1.0 $\cdot$ ckpt/250 steps $\cdot$ tested resume	fp16 spikes; Kaggle 12 h session cap
Prompt format	prompts end at the closing "" — no trailing newline	newline+indent is ONE BPE token; a bare <code>\n</code> says “dedent next” and the model closes the function. Removing one character: pass@10 went 0 $\rightarrow$ 0.4

Results — against real baselines

Criteria 2+3 · all numbers in results/metrics.json



Metric	Random init	GPT-2 124M	PyGPT (ours)
Val perplexity (Python)	38,487	n/a (other vocab)	6.19 (5.24 @ 1 epoch)
AST-valid samples	0%	7%	57%
pass@1 (5 DSA tasks)	—	0.00	0.09
pass@10 (5 DSA tasks)	—	0.00	0.58

Loss 10.4 → 1.639 over two epochs; a final DSA curriculum phase trades +0.18 nats of general loss for 1.5× the functional pass rate — specialization, measured.

pass@10 journey: 0.00 → 0.39 by removing ONE character from the prompt (the newline/BPE finding, slide 8) → 0.58 with the curriculum. GPT-2 stays at 0.00 throughout.

is_prime: 0 passes for every model — the honest boundary of 110M params at this budget.

Live demo

Criterion 4

- `python app/app.py` — runs offline on this laptop, ~2.4 s per completion (KV cache)
- Prompt A: `def fibonacci(n) + docstring` — read what it does, not what we hope it does
- Prompt B: `NumPy normalize()` — has it learned library idioms?
- Prompt C: `class Stack with __init__` — multi-method structure
- Same prompt at temperature 0.2 vs 0.9 — sampling in one sentence
- Prompt F: `solve_sudoku()` — the honest failure case
 - backups, in order: Kaggle public share link · recorded video · screenshots

Design ↔ implementation: nothing on these slides is vaporware

Criterion 4

Design component	File · symbol	Verified by
Data filtering + tokenization	data/prepare_data.py · keep_file, tokenize_stream	token counts re-counted from disk; decoded sample inspected
Causal self-attention	model.py · CausalSelfAttention	test: changing token $t+1$ never changes logits $\leq t$
Full model + init	model.py · GPT	init loss $10.42 \approx \ln(V)$; overfits one batch to < 0.1 ; 110.5M params
Training loop	train.py · get_lr, estimate_loss, save_ckpt	smoke run · kill + --resume continues · DDP on 2 GPUs · time-limit exit
Sampling + KV cache	model.py · generate	cached output bit-identical to uncached (sanity check 5)
Evaluation	eval.py · val_loss, ast_validity, functional_test	results/metrics.json — the numbers on slide 9
Export + demo	export.py, app/app.py	fp16 round-trip greedy-identical; this demo

Code: github.com/Rehan123-bash/pygpt · Weights + runnable demo: huggingface.co/rehannn11223/pygpt

Limitations and future work

Criterion 1 — knowing the boundary is part of the result

- Under-trained $\sim 3\times$ vs Chinchilla-optimal (0.8B of ~ 2.2 B tokens) — loss was still falling when we stopped
- Completion $\neq$ correctness: $\text{pass}@10 = 0$ on toy tasks; it autocompletes plausibly, it does not reason
- 512-token context — function-level, not file-level, completion
- No instruction tuning: “write a function that...” prompts will not work
- Specialization tax: the DSA curriculum cost +0.18 nats of general-Python loss — narrow gains aren't free
- Future work:
 - longer training — 22 unused free GPU-hours per week were left on the table
 - train our own tokenizer · fill-in-the-middle objective · VS Code inline-completion extension (backend already exists)

References

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- Rozière et al. 2023 — Code Llama
- Kaplan et al. 2020 — Scaling Laws · Hoffmann et al. 2022 — Chinchilla
- Sennrich et al. 2016 — BPE · Press & Wolf 2017 — Weight Tying
- Loshchilov & Hutter 2019 — AdamW · Micikevicius et al. 2018 — Mixed Precision
- Holtzman et al. 2020 — The Curious Case of Neural Text Degeneration (nucleus sampling)
- Karpathy — nanoGPT (reference implementation pattern)